

Development of a Novel Technique to Record 3D Intersegmental Angular Kinematics During Dynamic Spine Movements

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Abstract—A novel method of recording intersegmental spine kinematics was developed. The method was required to: (1) have similar accuracy and precision as current methods that record gross spine kinematics; (2) be reasonably insensitive to errors associated with marker detection or misplacement; and (3) be reasonably insensitive to skin movement artefacts. Four healthy participants performed trunk flexion, lateral bending, and axial twist movements; data were collected using the intersegmental method as well as electromagnetic sensors. Comparing methods, gross angular kinematic differences were within 1 SD during flexion and lateral bend, while axial twist resulted in the largest differences. To test sensitivity to marker error, random error was added to marker positions. The most proximal and distal intersegmental units were the most sensitive to marker error. Adding additional markers at the ends or interpolating padded markers reduced this sensitivity. The influence of skin movement artefact was investigated by digitizing locations of the skin with respect to the spinous processes in both neutral and fully flexed postures. In the lumbar region, large skin artefacts had minimal effect on intersegmental angles. The greatest strength of this method is the ability to dynamically record intersegmental spine kinematics.

Keywords—Biomechanics, Curvature, Electromagnetic Intervertebral, Lumbar, Motion, Movement, Multisegment, Thoracic, Model.

ABBREVIATIONS

AP	Anterior/posterior axis
AT	Cardan angle about the superior/inferior axis

C7	7th cervical vertebrae
FE	Cardan angle about the mediolateral axis
LB	Cardan angle about the anterior/posterior axis
LCS	Local coordinate system
S1	1st sacral vertebrae
ML	Mediolateral axis
SI	Superior/inferior axis
T12	12th thoracic vertebrae

INTRODUCTION

Intersegmental spine instability has long been considered a risk factor for the development of spine injury and back pain.^{7,11,18,23,26} Intersegmental instability is believed to cause instances of segmental spine buckling leading to high stresses and strains within spine tissues.^{9,24} This mechanism of injury is supported by a wide body of research studying the effects of large loads and displacements on functional spine units of human cadavers^{1,2,10,21} and animals.^{13,15,36} While excessive intersegmental motion and instability is generally agreed to be a mechanism of spine injury, detecting intersegmental kinematics during movement is very challenging.

The most common method of recording lumbar spine kinematic data is to calculate the relative transformation between two rigid bodies or sensors spanning the entire lumbar spine.^{14,33,34} These measures of gross spine movement can be collected using optical rigid bodies, electromagnetic sensors, or accelerometers. These techniques all track movement of the skin overlying the spine rather than the vertebrae themselves. Errors in marker placement, marker detection,

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and skin artefacts are all known to influence these measures.¹⁷ These sources of error are generally much smaller than kinematic signals from gross spine movement; however, they pose a problem for resolving smaller intersegmental angular motion.

Intersegmental kinematics can be obtained using medical imaging modalities such as CT, MRI, or fluoroscopy.^{5,29} CT and MRI have relatively high resolution and are able to create 3D spatial reconstructions of the vertebrae with high levels of accuracy. By directly imaging the vertebral structures, these methods are not influenced by marker errors or skin artefacts. However, these modalities are prohibitive as they are expensive, not readily available, and do not provide information during dynamic movements. In addition, most CT and MRI scans are taken in unloaded supine spine postures. One benefit of fluoroscopy is the ability to collect kinematic vertebral data during dynamic movements.^{3,6,35} However, fluoroscopy is generally limited to a single plane and yields smaller, lower resolution images.

Recently, there have been advancements for recording segmental spine kinematics using skin-mounted systems. A limiting factor is that most sensors and rigid bodies are too large to be placed over adjacent vertebrae.³⁰ Preuss and Popovic developed a novel method of creating seven local coordinate systems from triads of skin-mounted markers, each spanning three intervertebral segments.²⁷ Local coordinate systems were created using traditional methods often reserved for rigid bodies of three non-collinear points. This method yielded the highest spatial resolution of spine posture to date, tracking six regions spanning the thoracic and lumbar spine. Another approach developed by Cloud *et al.* fit an array of optical markers over the sagittal midline of the spine with a planar cubic spline.⁸ This method enabled measurement of intersegmental flexion angles based on vectors normal to the spline. A limitation of this method is that it is constrained to 2D analyses. If the spline representing the sagittal midline is mapped to 3D, there is no longer a unique normal vector to the curve; rather there is a normal plane for each point along the spline. This issue could be resolved by combining aspects of the methods used by Preuss and Popovic²⁷ and Cloud *et al.*⁸ The medio-lateral axis of the body can be defined by additional columns of markers lateral to the midline, similar to Preuss and Popovic. This axis can be used to define the vector normal to the 3D spline and perpendicular to the medio-lateral axis of the body.

Using these ideas, the purpose of this study was to develop a novel technique of recording intersegmental kinematics of the spine. The goal of developing this method was to collect 3D intersegmental spine kinematics during dynamic, complex movements in any

direction. The technique was required to meet a number of expectations: (1) The angular kinematics of the spine must be equivalent in accuracy and precision to currently acceptable methods of gross spine movement; (2) the technique must be minimally sensitive to marker errors which may arise from marker misplacement or marker detection; (3) the method must be minimally sensitive to skin movement artefacts.

MATERIALS AND METHODS

Intersegmental Method

Three columns of 21 passive markers (6 mm diameter reflective discs) were placed over the spine. The midline column was centered over the spinous processes from C6 to S2. The left and right columns were 3–5 cm lateral to the spinous processes, over the apex of the erector spinae muscles (Fig. 1a). Locations of reflective markers were recorded at 120 Hz (Optitrack, NaturalPoint Inc., Corvallis, OR, USA) and filtered with a zero-lag 4th order low-pass Butterworth filter with an effective cut-off frequency of 2 Hz.

The X, Y, and Z positions of each column of markers were fit with piecewise—five knots and six segments—cubic polynomials parameterized as a function of spine level. The result of this yielded a 3D spline for each of the left, midline, and right columns. Orthonormal local coordinate systems (LCS) were created at each spine level from C7 to S1. The origin of each LCS was defined by evaluating the midline spline. The superior/inferior (SI) vector of each LCS was tangent to the midline spline. The anterior/posterior (AP) vectors were perpendicular to the SI vectors and the vectors connecting the left and right splines. The vectors connecting the left and right splines created a 3D surface, which can be considered the anatomical frontal plane. The mediolateral (ML) vectors were perpendicular to the SI and AP vectors (Fig. 2).

Part 1: Comparison Between Electromagnetic and Intersegmental Methods

Four healthy male participants (mean $\pm$ SD; age: 27 ± 1.7 years; height: 1.8 ± 0.1 m; mass: 85 ± 10.3 kg) with no history of low back pain were recruited with approval from the local research ethics board and provided consent for participation in the study.

Spine kinematic data were collected using the intersegmental method outlined above as well as using three electromagnetic sensors (Polhemus Liberty, Colchester VT, USA) attached to rigid plates (3×10 cm) adhered to the skin over the C7, T12, and

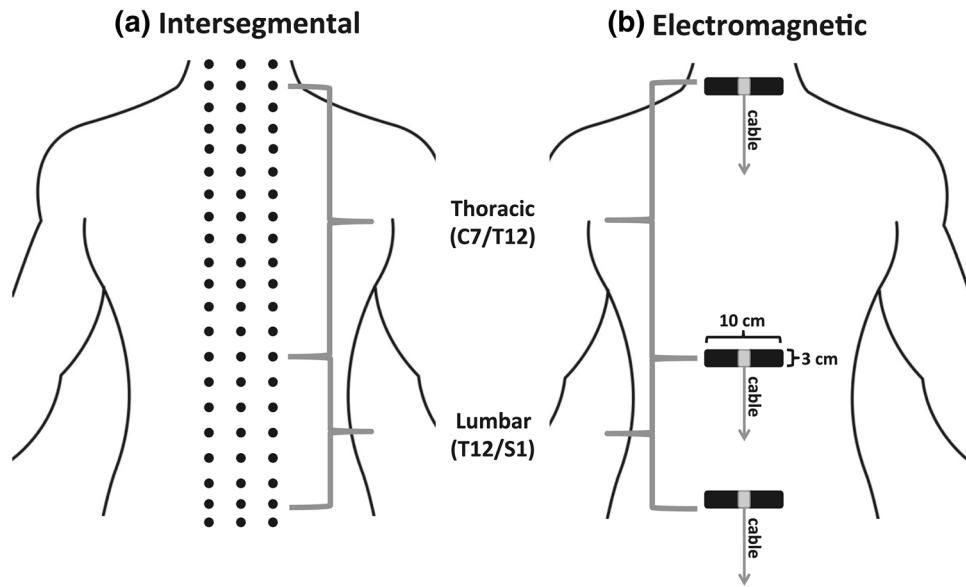


FIGURE 1. Methodological figure depicting setup for (a) the intersegmental method and (b) the electromagnetic method.

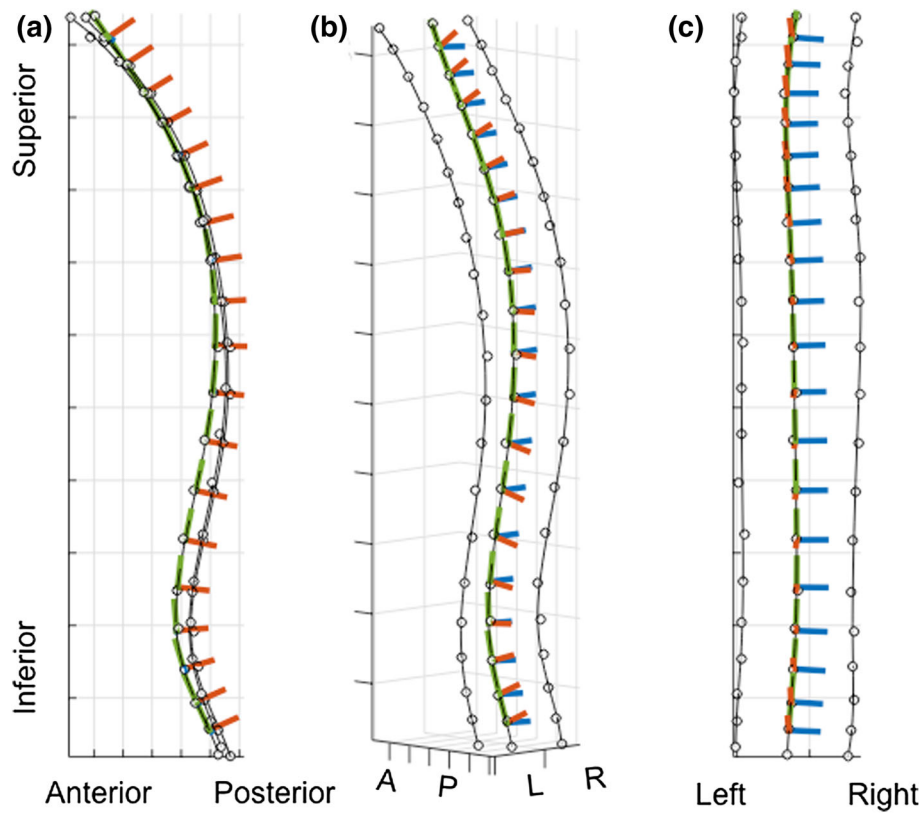


FIGURE 2. Spatial reconstruction of the intersegmental method. (a) Sagittal view, (b) posterolateral view, and (c) posterior view of one participant in neutral standing. Open white circles are passive markers and black curves are left, midline, and right 3D splines. Local coordinates systems are shown for each spine level from C7 to S1. SI vectors (green) are tangential to midline 3D spline. AP vectors (red) are perpendicular to the surface containing the left and right 3D splines and the SI vectors. ML vectors (blue) are perpendicular to SI and AP vectors.

S1 spinous processes (Fig. 1b). Rigid plates were used because pilot testing with sensors directly attached to the skin caused errors during axial twist (bulging of erector spinae muscles twisted the sensors in the opposite direction of the movement). An electromagnetic system was used for comparison as these sensors are accurate, reliable, and are widely accepted for tracking spine movement.^{16,19,20,22,25,31} The positions and orientations of the electromagnetic sensors were sampled at 120 Hz and filtered with a zero-lag 4th order low-pass Butterworth filter with an effective cut-off frequency of 2 Hz.

Participants performed self-paced targeted flexion, lateral bend, and axial twist movements of the spine. Each movement was repeated six times—three times with the electromagnetic technique and three times with the intersegmental technique. Each participant's pelvis was constrained against a stable support to ensure spine rather than hip movement.^{12,28} Participants began and ended all trials in a neutral standing position. For flexion trials, participants were instructed to flex through their spine with arms held across their chest and touch their forehead to a target (0.45 m anterior and 1.35 m from the floor). For lateral bend trials, participants bent laterally until their fingertips touched their fibular head; participants bent to their right then to their left. For axial twist trials, participants held their arms across their chests and axially twisted to visual targets (0.45 m anterior and 0.33 m lateral). Participants twisted to the target on their right then to the target on their left.

For both methods, three dimensional lumbar and thoracic spine angles were obtained by calculating the Cardan angles of T12 relative to S1 and C7 relative to T12, respectively. The same flexion–extension (FE), lateral bend (LB), axial twist (AT) Cardan sequence was used to calculate relative angles for all trials. Intersegmental angles were also computed from C7/T1 through to L5/S1 using data collected with the intersegmental method and the same Cardan sequence as above.

Movement trials were time-normalized to account for different speeds of movement between trials and participants. The electromagnetic and intersegmental methods were compared by analyzing the average and standard deviation of each Cardan angle (FE, LB, and AT) for each movement (flexion, lateral bend, and axial twist) for both lumbar and thoracic spine regions.

Part 2: Sensitivity to Marker Error

It was expected that the 3D splines would be the least constrained at the most superior and inferior spine levels; therefore, these intersegmental angles would be the most sensitive to errors caused by marker

placement and marker tracking. Two different methods of constraining the 3D splines were considered to reduce the sensitivity to marker error. First, additional markers could be placed superiorly and inferiorly on the participant. This was the method used in Part 1 where C6 and S2 marker positions were recorded, however LCSs were not constructed at these levels. Second, the superior and inferior ends of the spline could be padded by creating virtual markers. The positions of these virtual markers were obtained by linearly extrapolating from the two most superior and inferior markers.

Sensitivity analyses were conducted using four different test conditions to identify which method would be best for handling spline instability. The four conditions were: (1) no additional markers (No Extra); (2) additional markers on the participant at C6 and S2 spine levels (Extra); (3) three virtual markers were created and padded both the superior and inferior ends of each spline (Padded); (4) additional markers on the participant at C6 and S2 and two virtual markers at both the superior and inferior ends of each spline (Extra + Padded).

The sensitivity of each condition was tested in four different postures (neutral, flexed, bent, and twisted). One thousand simulations were created in each posture by adding random error to X, Y, and Z positions of each marker. Errors were randomly sampled from a normal distribution with $\mu = 0$ and $\sigma = 0.762$ mm. The standard deviation (σ) was double the mean 3D reprojection error of the calibration trials (Motive, NaturalPoint, Corvallis, OR, USA). Therefore, based on the empirical rule, 27% of markers were displaced 2–4 times further from their true location than would realistically occur due to potential marker tracking errors.

The root mean square errors were compared intersegmentally for each Cardan angle (FE, LB, and AT) in order to test for differences in sensitivity to random marker noise.

Part 3: Sensitivity to Skin Movement Artefact

The error introduced due to relative movement between the skin and underlying vertebrae was investigated in full trunk flexion. Full flexion was chosen as it was believed to cause the greatest relative movement between the skin and vertebrae.

The skin over the spinous processes was marked in both neutral standing and full trunk flexion in the same four participants from Part 1. The displacements of the skin relative to the spinous processes were measured manually with a cloth tape measure in full trunk flexion and expressed as a percent of the intersegmental distance.

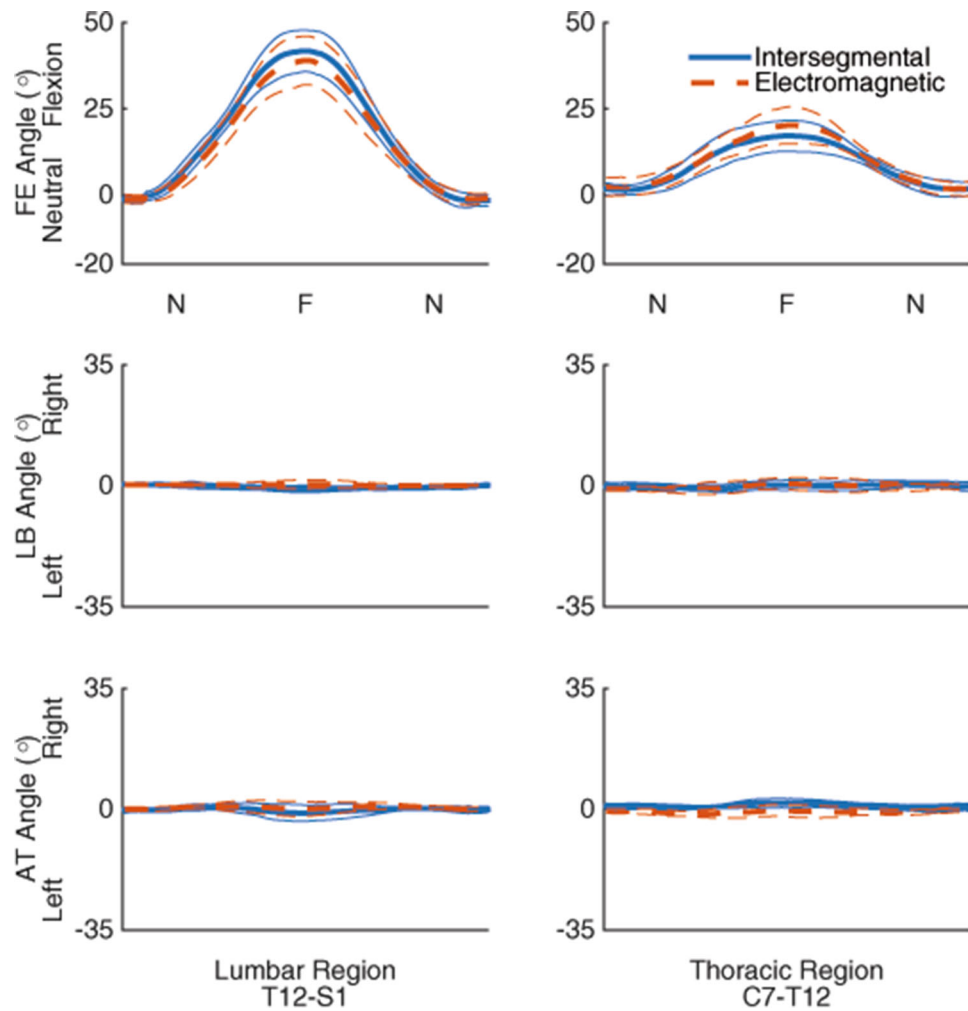


FIGURE 3. Time-normalized average (thick lines) $\pm$ 1 SD (thin lines) Cardan angles during targeted trunk flexion recorded using electromagnetic sensors (red dashed) and the novel intersegmental technique (blue continuous). Lumbar and thoracic angles are the relative rotations of T12 with respect to S1 and C7 with respect to T12, respectively. Vertical scaling between all panels is consistent for ease of comparison. *N* neutral posture, *F* targeted flexion.

In the target flexion postures from Part 1, locations of spinous processes were estimated based on the relative skin displacements measured in full flexion. LCSs were created at both the uncorrected marker locations and estimated spinous process locations. Skin movement errors were calculated as the difference between the uncorrected and estimated intersegmental angles. Note that errors due to relative skin movement were exaggerated because the relative displacements were calculated in fully flexed postures and applied to the submaximally flexed postures in Part 1. Therefore, the presented magnitudes of skin movement error are likely beyond the upper limit that would be observed in submaximally flexed postures.

RESULTS

Part 1: Comparison Between Electromagnetic and Intersegmental Methods

Presented data are mean $\pm$ SD. There was excellent agreement between the electromagnetic and intersegmental methods for all lumbar and thoracic angles during the trunk flexion movement (Fig. 3). At target flexion, the lumbar FE angles for the electromagnetic and intersegmental methods were $39.0 \pm 7.0^\circ$ and $41.8 \pm 6.0^\circ$, respectively. At target flexion, the thoracic FE angles for the electromagnetic and intersegmental methods were $20.2 \pm 5.3^\circ$ and $17.1 \pm 4.5^\circ$, respectively. The Off-Axis (LB and AT) angles throughout

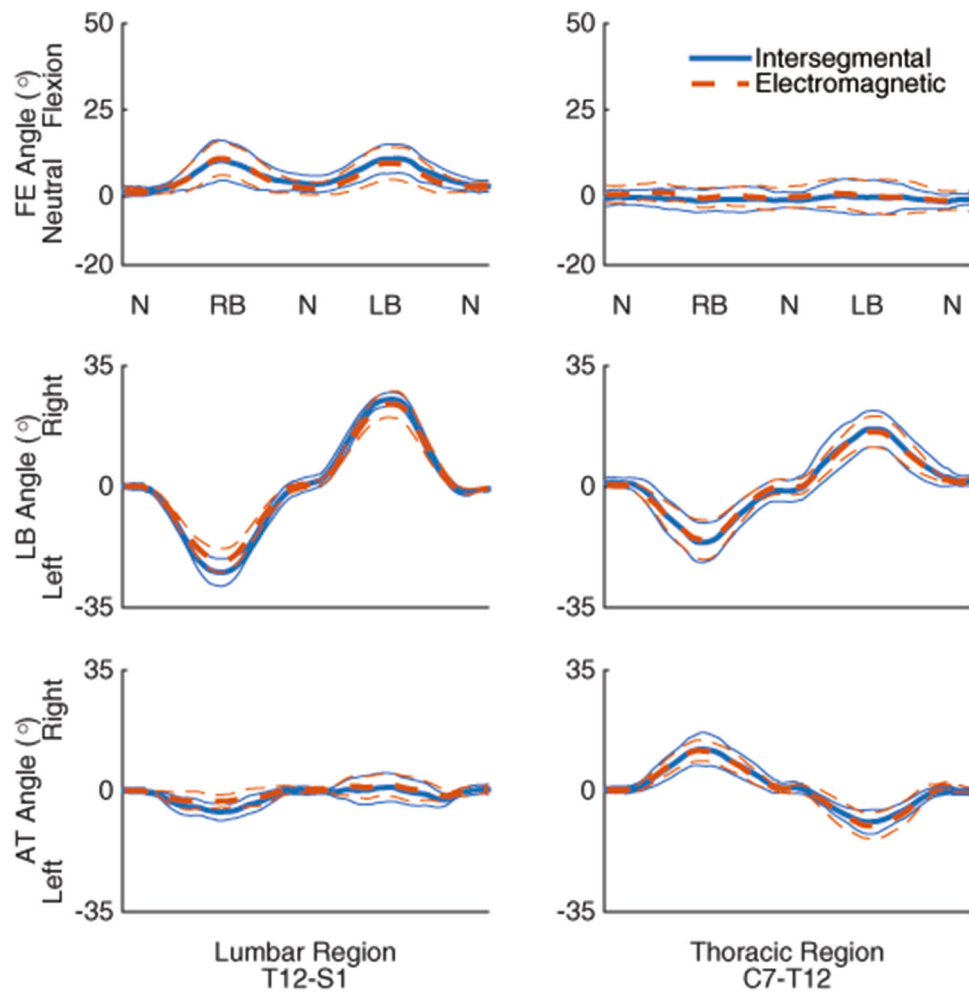


FIGURE 4. Time-normalized average (thick lines) ± 1 SD (thin lines) Cardan angles during targeted trunk lateral bend recorded using electromagnetic sensors (red dashed) and the novel intersegmental technique (blue continuous). Lumbar and thoracic angles are the relative rotations of T12 with respect to S1 and C7 with respect to T12, respectively. Vertical scaling between all panels is consistent for ease of comparison. *N* neutral posture, *RB* targeted right bend, *LB* targeted left bend.

the entire movement were $< 2.0^\circ$ for lumbar and thoracic regions recorded using both methods.

There was strong agreement in magnitudes and directions of lumbar and thoracic angles recorded using both methods during the lateral bending movement (Fig. 4). At targeted lateral bend, the electromagnetic and intersegmental methods detected lumbar LB angles (pooled across both bending directions) of $22.6 \pm 3.9^\circ$ and $25.1 \pm 3.2^\circ$, respectively. The thoracic LB angles detected with electromagnetic and intersegmental methods were $15.6 \pm 5.3^\circ$ and $16.4 \pm 5.6^\circ$, respectively. Both electromagnetic and intersegmental methods detected lumbar FE angles of $10.2 \pm 4.9^\circ$ and $10.6 \pm 5.2^\circ$ and small lumbar AT angles of $2.2 \pm 2.8^\circ$ and $4.0 \pm 4.3^\circ$, respectively in the same direction (i.e., left twist during left bend). In the thoracic region, neither electromagnetic nor intersegmental methods detected FE angles ($-0.3 \pm 4.0^\circ$ and $-0.9 \pm 4.3^\circ$, respectively); both methods detected AT angles

($10.8 \pm 3.5^\circ$ and $10.0 \pm 4.6^\circ$) in the opposite direction of bending (i.e., right twist during left bend).

The weakest agreement between the electromagnetic and intersegmental methods was during the axial twist movement (Fig. 5). AT angles were in close agreement in the lumbar region for both electromagnetic and intersegmental methods ($3.6 \pm 2.4^\circ$ and $3.3 \pm 2.4^\circ$, respectively). Lumbar FE angles during axial twist were $0.5 \pm 1.5^\circ$ and $0.3 \pm 1.0^\circ$ for electromagnetic and intersegmental methods, respectively. The intersegmental method detected greater lumbar LB angles during targeted axial twist than the electromagnetic method ($8.8 \pm 3.7^\circ$ and $2.8 \pm 1.7^\circ$, respectively). LB angles were contralateral to the direction of twist for both methods. In the thoracic region, the intersegmental method detected lesser AT angles (intersegmental: $11.7 \pm 4.4^\circ$; electromagnetic $16.4 \pm 5.4^\circ$) and greater LB angles (intersegmental: $15.9 \pm 5.4^\circ$; electromagnetic: $10.7 \pm 4.1^\circ$) than the electromagnetic method. The

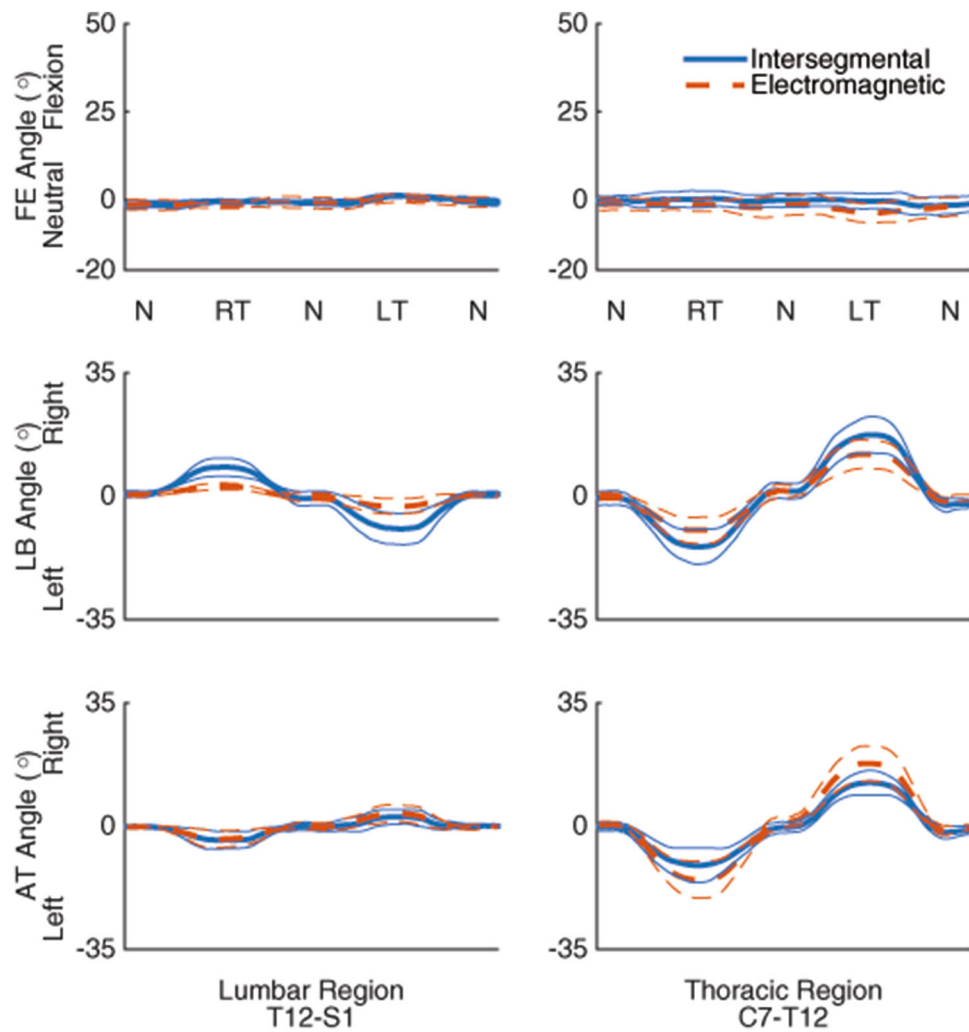


FIGURE 5. Time-normalized average (thick lines) ± 1 SD (thin lines) Cardan angles during targeted trunk axial twist recorded using electromagnetic sensors (red dashed) and the novel intersegmental technique (blue continuous). Lumbar and thoracic angles are the relative rotations of T12 with respect to S1 and C7 with respect to T12, respectively. Vertical scaling between all panels is consistent for ease of comparison. *N* neutral posture, *RT* targeted right twist, *LT* targeted left twist.

thoracic LB and AT angles were in the same direction. The FE angles during targeted AT were $-2.7 \pm 2.6^\circ$ (negative indicating thoracic extension) for the electromagnetic method and $-0.1 \pm 2.3^\circ$ for the intersegmental method.

Intersegmental angles for each participant in targeted flexion, lateral bend, and axial twist are shown (Fig. 6). Targeted movements were highly consistent within individuals evidenced by the small SD between the three trials. These four healthy participants with no previous history of low back pain exhibited different intersegmental movement patterns in all three movements.

Part 2: Sensitivity to Marker Error

Sensitivity to random error was independent of posture; therefore, all four postures were pooled for the presentation of results in this section.

The LB and FE angles were sensitive to random error in the No Extra condition. The C7/T1 and L5/S1 intersegmental angles had the greatest RMS errors (1.99 – 3.05°) and were the most sensitive to random error. The T1/T2 and L4/L5 intersegmental angles had RMS errors between 1.21 and 1.28° . The intersegmental angles between T2/T3 and L3/L4 were insensitive to error (all RMS errors $\leq 0.77^\circ$).

Adding one row of additional markers to the participant or padding the splines with virtual markers improved the sensitivity to random error compared to the No Extra condition. The Extra condition was marginally less sensitive to noise than the Padded condition as RMS errors from T1/T2 to L4/L5 were $\leq 0.58^\circ$ and $\leq 1.23^\circ$, respectively. C7/T1 and L5/S1 had RMS errors of 1.23 – 1.28° (Extra) and 0.5 – 1.74° (Padded).

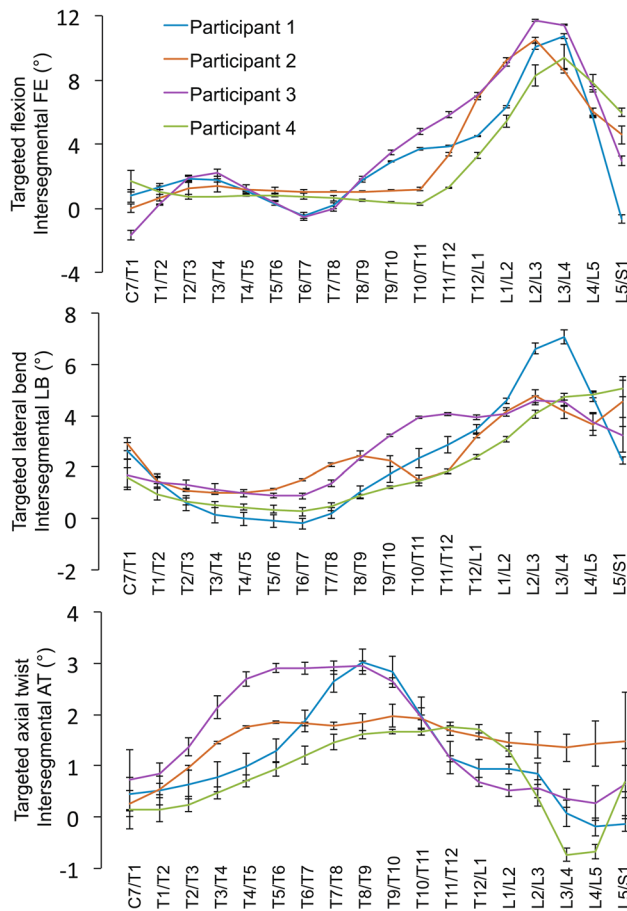


FIGURE 6. Intersegmental angles (mean $\pm$ SD) of each participant in targeted flexion, targeted lateral bend, and targeted axial twist. Differences in segmental movement patterns are observed between participants using the intersegmental method.

The combination of one row of additional markers and two rows of padded virtual markers proved the least sensitive to error. RMS errors of all intersegmental angles were $\leq 0.39^\circ$ (Extra + Padded).

In general, AT angles were less sensitive to marker errors than FE and LB angles (Table 1), and AT angles were equally sensitive across all spine levels. For AT angles the conditions from most sensitive to least sensitive to marker errors were No Extra > Padded > Extra > Extra + Padded.

Part 3: Sensitivity to Skin Movement Artefact

In a maximally flexed posture, relative to the underlying spinous processes the skin in the thoracic region stretched superiorly while in the lumbar region the skin stretched inferiorly (Fig. 7). The greatest skin displacements as a percent of intersegmental distance occurred in the upper thoracic region (Fig. 7). In some instances the skin over the C7, T1, and T2 spinous processes in neutral standing was over the C6, C7, and T1 spinous processes in the fully flexed posture.

While the relative skin displacement followed a relatively simple trend (Fig. 7), the same pattern was not observed in the skin movement errors of the intersegmental angles (Table 2). In the lumbar region (T12/L1 to L5/S1), relative skin displacement of -6 to -46% had minimal effect on the FE intersegmental angle error (average error: -0.4 to 0.1° ; minimum: -0.8° ; maximum: 0.7°). In the thoracic region, FE intersegmental angle errors due to skin artefact were greater with average errors ranging from -1.5 to 1.9° .

TABLE 1. RMS Errors (degrees) of intersegmental Cardan angles due to random marker error.

	No Extra			Extra			Padded			Extra+Padded		
	AT	LB	FE	AT	LB	FE	AT	LB	FE	AT	LB	FE
C1/T1	0.87	1.99	2.01	0.29	1.24	1.26	0.93	1.74	1.28	0.29	0.36	0.36
T1/T2	0.33	1.28	1.28	0.34	0.46	0.47	0.66	0.75	1.23	0.21	0.38	0.38
T2/T3	0.42	0.45	0.44	0.33	0.42	0.41	0.29	0.48	1.06	0.13	0.34	0.34
T3/T4	0.30	0.76	0.77	0.19	0.58	0.58	0.18	0.29	0.77	0.17	0.19	0.19
T4/T5	0.23	0.57	0.57	0.24	0.30	0.31	0.24	0.13	0.40	0.20	0.11	0.11
T5/T6	0.30	0.22	0.22	0.26	0.18	0.18	0.22	0.20	0.21	0.17	0.18	0.18
T6/T7	0.25	0.39	0.39	0.19	0.38	0.37	0.16	0.32	0.60	0.12	0.27	0.27
T7/T8	0.19	0.48	0.48	0.16	0.34	0.34	0.15	0.21	0.48	0.14	0.18	0.18
T8/T9	0.26	0.20	0.20	0.23	0.15	0.14	0.18	0.10	0.24	0.17	0.09	0.09
T9/T10	0.25	0.21	0.22	0.23	0.16	0.16	0.17	0.11	0.11	0.17	0.10	0.10
T10/T11	0.18	0.40	0.41	0.17	0.30	0.30	0.14	0.18	0.27	0.14	0.17	0.17
T11/T12	0.25	0.27	0.27	0.19	0.26	0.26	0.13	0.22	0.40	0.12	0.20	0.20
T12/L1	0.30	0.16	0.16	0.26	0.12	0.12	0.18	0.13	0.23	0.17	0.12	0.12
L1/L2	0.23	0.40	0.40	0.24	0.21	0.20	0.20	0.09	0.09	0.20	0.07	0.07
L2/L3	0.34	0.58	0.60	0.21	0.46	0.44	0.18	0.18	0.24	0.18	0.14	0.14
L3/L4	0.42	0.35	0.36	0.37	0.39	0.40	0.21	0.36	0.48	0.15	0.28	0.29
L4/L5	0.31	1.21	1.23	0.33	0.41	0.41	0.36	0.48	0.52	0.25	0.37	0.39
L5/S1	0.93	2.97	3.05	0.28	1.28	1.23	0.39	0.52	0.50	0.31	0.37	0.37

Error magnitudes are associated to the colour intensity in each cell.

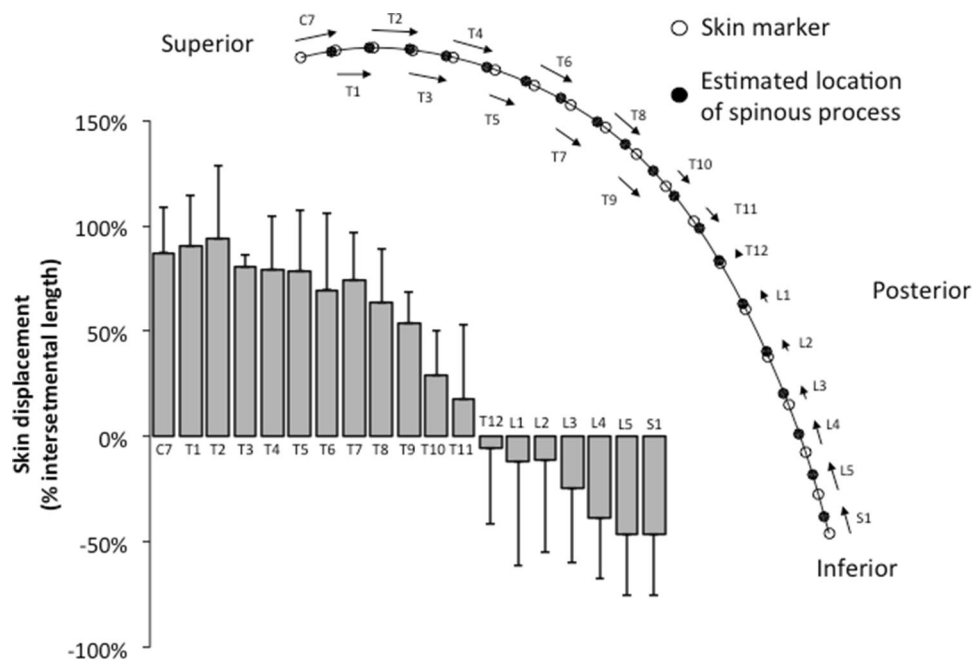


FIGURE 7. Sagittal view of midline spline in targeted flexion posture. Open circles are the recorded marker locations and black circles are the estimated spinous processes locations. Inset is the segmental relative skin displacement (mean $\pm$ SD) of all four participants calculated in full flexion.

TABLE 2. Segmental errors (degrees) of intersegmental Cardan angles due to relative skin movement.

	Mean	SD	(Min to Max)
C1/T1	-0.9	0.4	(-1.7 to -0.4)
T1/T2	-0.2	1.1	(-1.5 to 1.6)
T2/T3	2.2	0.9	(0.8 to 3.3)
T3/T4	0.3	0.4	(-0.5 to 0.7)
T4/T5	0.0	0.2	(-0.2 to 0.3)
T5/T6	0.8	0.5	(0.0 to 1.5)
T6/T7	-1.2	0.6	(-2.3 to -0.8)
T7/T8	0.3	0.2	(0.0 to 0.6)
T8/T9	0.2	0.4	(-0.2 to 0.9)
T9/T10	1.7	0.7	(1.2 to 2.8)
T10/T11	0.6	0.6	(0.0 to 1.5)
T11/T12	1.6	0.6	(0.5 to 2.4)
T12/L1	0.0	0.2	(-0.2 to 0.3)
L1/L2	0.2	0.2	(0.0 to 0.6)
L2/L3	0.3	0.2	(0.1 to 0.6)
L3/L4	0.2	0.7	(-0.9 to 0.9)
L4/L5	-0.1	0.1	(-0.3 to 0.0)
L5/S1	-0.2	0.4	(-0.7 to 0.4)

Mean, SD, minimum, and maximum are from all twelve targeted flexion postures (4 participants $\times$ 3 repetitions).

DISCUSSION

Overall the novel method described in this paper is believed to accurately and reliably detect 3D intersegmental motion of the spine. The method agreed with electromagnetic tracking of spine regions, was insensitive to random marker noise, and moderately

affected by skin movement artefacts. The greatest advantage of the intersegmental technique is the increased spatial resolution (Fig. 6). Increasing the spatial resolution enables the detection of differences in spine movement patterns in healthy individuals, which cannot be observed by gross kinematic measures.

Agreement Between the Intersegmental and Electromagnetic Methods

The level of agreement between the novel intersegmental technique and electromagnetic tracking of spine regions ranged from excellent during trunk flexion to good during axial twist. Notable differences between the methods are highlighted below.

As expected, both methods detected planar spine movement during trunk flexion. The magnitudes and directions of all three Cardan angles from both systems were in close agreement. The data therefore suggest that the intersegmental method is as accurate at recording gross trunk movement as the conventionally used electromagnetic system during dynamic trunk flexion.

Both the electromagnetic and intersegmental methods were also in close agreement during lateral bending. The intersegmental method detected slightly greater LB range of motion than the electromagnetic method due to differences in skin movement. Lateral bending to the left tensions the skin on the right side of the spine. The intersegmental method did not restrict skin movement and this tensioning was detected as a greater LB angle. The electromagnetic sensor was secured with a rectangular plate taped over the left and right erector spinae muscles. This rigid plate helped ensure that the sensor was not affected by muscle bulging; however it also limited the amount of natural skin stretch and subsequently the LB angle. In the absence of a true gold standard for the recording of vertebral motion, it is difficult to determine which method most closely matches the underlying vertebrae. Due to the relatively small differences the methods may be considered equivalent in regard to precision and accuracy.

The greatest differences between the two methods were observed during axial twisting. The intersegmental method detected 6° and 4° greater LB angles in the lumbar and thoracic regions, respectively, and 5° lesser thoracic AT angles. These differences are likely caused by the skin translating laterally over the spinous processes when using the intersegmental method. Attaching the electromagnetic sensor *via* plates helped to restrict lateral skin translation; however, as seen during the lateral bending movement, these plates also reduced the magnitude of the LB angles. Radiographs taken during axial twist movements have shown that some lateral bend does occur in the lumbar spine during axial twist movements.³² This corresponds more closely with the results of the intersegmental rather than the electromagnetic method in the lumbar region.

Sensitivity to Marker Error

When using optical systems, marker errors are expected due to misplacement of markers and errors in

software tracking. These errors are less problematic when tracking spine kinematics of the entire lumbar or thoracic spine because kinematic signals are considerably greater than these errors. However, intersegmental angles are smaller and may be more susceptible to low signal to noise ratios. The simplest method of collecting intersegmental kinematics would be to increase the spatial density of skin markers and apply similar techniques as Preuss and Popovic.²⁷ A difficulty of this is that as the distance between markers is reduced, local coordinate systems created using skin mounted markers become increasingly susceptible to noise. Further, the surface over each vertebral segment is small, thereby limiting the number of markers that can be clustered. In the current study, this issue was overcome by fitting the markers with piecewise cubic spline polynomials. Since polynomials are less constrained at the ends, it was important in Part 2 to determine how sensitive the ends of the spline were to marker error and whether this sensitivity was posture dependent.

As expected, the greatest sensitivity to marker noise existed at the superior and inferior ends of the 3D curves. This increased sensitivity only affected the FE and LB angles. AT angles were the least sensitive to marker error, because rotations about this axis were constrained by the right and left columns of markers. Posture also did not influence the sensitivity to marker noise. This is beneficial because users do not need to be concerned about increased marker error at end ranges of spine motion.

As expected, adding additional markers or padding the spline with extrapolated virtual markers at the superior and inferior ends both decreased the sensitivity to marker error; however, there was still increased sensitivity at the most superior and inferior intersegmental angles. Adding one additional marker to the participant was slightly better than padding the data with three virtual markers. However, when adding additional markers to the participant is not feasible, padding three layers of virtual markers is acceptable and preferred over not adding any additional or padded markers. In the future, adding one additional layer of markers and padding two layers of virtual markers may be the best method of decreasing sensitivity to marker error.

Sensitivity to Skin Movement Artefact

The skin is required to stretch and translate relative to the underlying vertebrae during normal movement. This movement between the skin and vertebrae influences all skin-mounted kinematic measures.²⁵ The inertia of large sensors and rigid plates is believed to further influence kinematic measures. Depending on

the type and speed of movement, the sensor inertia may increase or decrease the movement of the skin relative to the vertebrae. In the current study, the low-mass markers did not restrict or artificially alter skin movement; therefore, any skin movement that normally exists would be present in the data. Part 3 of this study aimed to determine how skin movement during trunk flexion influenced the calculation of intersegmental spine angles.

Re-digitizing the spinous processes in a flexed spine posture revealed substantial skin movement in both lumbar and upper thoracic regions. This artefact had minimal effect on the intersegmental angles in the lumbar region because this region had very little curvature in a flexed posture. Evaluating the LCSs at the estimated spinous process locations minimally changed their orientation. In the thoracic region, the smaller distances between vertebral levels, and greater curvature, had a larger effect on the intersegmental angles. Even though the relative skin movements were fairly consistent between thoracic levels and participants (Fig. 6), the effect on intersegmental errors varied between segments and participants (Table 2). This was due to different segmental movement patterns between participants and thoracic spinal curvatures.

A limitation of Part 3 of this analysis is that only skin movement artefacts during trunk flexion were considered. This was because trunk flexion has been shown to generate significant skin deformations^{4,17,37} and is believed to cause the greatest magnitudes of skin movement artefacts. However, lateral bend and axial twist also cause the skin over the spinous process to translate laterally and the effect of this translation on the calculation of angular kinematics is unknown. Because the direction of translation is unknown (as opposed to during flexion when the direction is along the sagittal midline), these effects could not be similarly modelled. A future study involving a full 3D surface map of the spine during lateral bend and axial twist will be needed to determine the effect of lateral skin artefacts.

CONCLUSION

Intersegmental spine kinematics can be collected during complex dynamic tasks from skin mounted passive markers. The novel method presented in this paper had similar levels of accuracy and precision as a traditionally used electromagnetic system. The method was also relatively insensitive to marker noise. Similar to other skin mounted techniques, skin artefacts influenced the detection of intersegmental kinematics. The greatest strength of this method is the increased spatial resolution of spine kinematics, which can be

used to detect intersegmental motion patterns, stiffness, and stability.

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